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A reliability estimation approach via Wiener degradation model with measurement errors

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ABSTRACT

This paper proposes a reliability estimation approach based on EM algorithm and Wiener processes by considering measurement errors. Firstly, the time-transformed Wiener processes are used to model the degradation process of the product, which simultaneously consider the temporal variability, unit-to-unit heterogeneity and measurement errors. In addition, we obtain the closed-form expressions of some reliability quantities such as reliability function and probability density function of the life. Moreover, the expectation maximization algorithm is adopted to estimate the model parameters effectively. Finally, a numerical example and a practical case study for LED lamps are provided to illustrate the effectiveness and superiority of the presented approach.

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1. Introduction

Reliability estimation based on degradation data has attracted a great deal of interest in reliability analysis and maintenance scheduling for highly reliable products [1–4]. Degradation data can provide much more reliability information for deteriorating products than the failure data because they can characterize the underlying failure processes [5–7]. Compared with the traditional failure data analysis, degradation analysis are often able to obtain more accurate reliability estimation under the circumstance of less testing units. The degradation evolution of many products arises in a stochastic way such as LED lamps, rotating machines, and bridge beams. Therefore, the use of stochastic processes can be helpful to effectively model the degradation process of the product and estimate the reliability quantities. Based on the reliability estimation results, preventive maintenance scheduling can be timely triggered to minimize the unscheduled downtime and reduce the cost of the whole life cycle [8,9].

The stochastic processes with independent increments, such as Wiener processes and inverse Gaussian processes, have been widely used in the degradation modeling of products. For the non-monotonic degradation process frequently encountered in practice, Wiener processes are a more appropriate choice since they not only have some favorable mathematical properties but also can model the non-monotonic degradation evolution. Whitmore and Schenkelberg in [10] applied a regular Wiener process to model the degradation process of a self-regulating heating cable. Tseng et al. in [11] further developed an integrated Wiener process to model the degradation process of LED lamps. These works only considered the effect

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of the temporal variability associated with the degradation evolution of a product. However, the degradation processes of many products are often affected by the unit-to-unit heterogeneity and measurement errors. The unit-to-unit heterogeneity means that the degradation paths among the same batch of products can be quite different owing to certain discrepancy in raw materials or manufacturing processes, and the observed degradation data are generally contaminated by measurement errors because of imperfect instruments and random environments.

In the existing degradation modeling works based on Wiener processes, the unit-to-unit heterogeneity has been well characterized by assuming certain parameters in the degradation model follow certain distributions. Peng and Tseng in [12] presented a degradation modeling approach based on the Wiener process with linear drift by considering measurement errors. To capture the unit-to-unit heterogeneity among a bath of products, they assumed the drift parameter follows a normal distribution, and obtained an analytically tractable degradation model. Nevertheless, it is noteworthy that the effect of measurement errors on the life distribution was not considered in their approach, therefore the obtained life estimation results are approximation. Si et al. in [13] applied the same degradation model to characterize the degradation process of the gyro in an inertial navigation platform. This work took the effect of measurement errors on the life distribution into account, but ignored the effect of the unit-to-unit heterogeneity on the life estimation. Ye et al. in [14] used a time-transformed Wiener process model with measurement errors to analysis the wear problem of magnetic heads used in hard disk drives and LED degradation problem, in which the normal distribution is also used to describe the unit-to-unit heterogeneity. However, it is worth noting that the expresses of the life distribution are not given in their work. The closed-form expresses of the life distribution are quite valuable for support decision-related applications such as maintenance scheduling, logistic planning, etc. For these reasons, it is necessary to simultaneously consider the temporal variability, unit-to-unit heterogeneity and measurement errors to provide more accurate reliability estimation results for subsequent maintenance scheduling. In addition, the above works obtain the maximum likelihood estimation (MLE) of the model parameters only by the direct maximization of the likelihood function. However, the estimated results are unsatisfactory when considering unit-to-unit heterogeneity in the degradation modeling. The main reason lies in the fact that the likelihood function includes unobserved hidden variables. As such, we need to investigate more effective parameter estimation method to obtain more accurate reliability estimation.

Motivated by the aforementioned works, the objective of this paper is to develop a reliability estimation approach by using a time-transformed Wiener process with measurement errors. The main contributions of this paper are summarized as follows. Firstly, the temporal variability, unit-to-unit heterogeneity and measurement errors are simultaneously taken into account in the developed degradation modeling method. Secondly, we obtain an exact yet closed-form expressions of the reliability quantities, such as reliability function and probability density function of the life, which can provide valuable information for subsequent maintenance scheduling. Thirdly, the expectation maximization (EM) algorithm is effectively used to obtain the MLE of the unknown parameters since the likelihood function contains unobserved hidden variables.

The rest of this paper is organized as follows. Section 2 presents the degradation modeling principle based on a time-transformed Wiener process with measurement errors and the closed-form expression of the reliability quantities in terms of the presented degradation model, while the statistical inference based on the EM algorithm is developed in Section 3. Section 4 provides a practical case study to illustrate the effectiveness of the proposed approach. Section 5 concludes this paper.

2. Degradation modeling and reliability estimation

Let $X(t)$ represent the underlying degradation evolution of a deteriorating system over time t, which is widely modeled by Wiener processes [15–17]. As a general rule, the underlying degradation state cannot be observed directly due to the inaccuracy of the measurement. Therefore, the observed degradation data are often influenced by imperfect measurement, i.e., the measurement process of $X(t)$ includes the measurement errors. To characterize the influence of the measurement errors, the observed degradation process can be expressed as

$$Y(t) = X(t) + \varepsilon = X(0) + \beta\Lambda(t) + \sigma W(\Lambda(t)) + \varepsilon, \tag{1}$$

where β and σ represent the drift parameter and diffusion parameter respectively. $\Lambda(t)$ denotes the transformed time scale, which is generally a monotone function with $\Lambda(0)=0$. $W(\Lambda(t))$ represents a nonstandard Brown motion process in the case of $\Lambda(t)\neq t$, which describes the temporal uncertainty of the degradation process. ε denotes the measurement error, which is generally assumed to be normally distributed with zero mean and standard deviation σ_ε. In addition, ε is assumed to be s-independent with β. Without loss of generality, the initial degradation state $X(0)$ is assumed to be zero in the following. In the degradation model above, the drift parameter β generally represents the degradation rate of a system, and the diffusion parameter η has no definite physical meaning. Considering the randomness of the materials and operating conditions, there always exist the different degradation rates for diverse systems in engineering practice. As such, it is more suitable to incorporate unit-to-unit heterogeneity into the degradation modeling. A typical assumption is that the drift parameter β is regarded as a random parameter for describing unit-to-unit heterogeneity, and σ is considered to be a fixed parameter for depicting the degradation characteristic common to all systems within the population. In general, the drift parameter β is supposed to follow a normal distribution, which has been widely adopted by many studies [14,18,19].

In the following, we will illustrate how to obtain the reliability quantities in terms of the degradation model (1). A system is often regarded as a failure when the degradation path first reaches a predefined failure threshold ω. Therefore, the concept of the first hitting time (FHT) is used to define the life. Without loss of generality, we assume the degradation path of a system increases over time. For the degradation process without the measurement errors, based on the FHT concept, the life T can be defined as

$$T = \inf\{t : X(t) \geq \omega | X(0) < \omega\}. \tag{2}$$

According to the property of the Wiener process, $\Lambda(T)$ follows the inverse Gaussian distribution when the drift parameter β is fixed [2]. Therefore, the reliability function is expressed by

$$R_{T|\beta}(t) = P(T \geq t) = P(Y(t) \leq \omega) = \Phi\left(\frac{\omega - \beta\Lambda(t)}{\sigma\sqrt{\Lambda(t)}}\right) - \exp\left(\frac{2\beta\omega}{\sigma^2}\right)\Phi\left(-\frac{\omega + \beta\Lambda(t)}{\sqrt{\sigma^2\Lambda(t)}}\right), \tag{3}$$

and the probability density function (PDF) of the life T is given by

$$f_{T|\beta}(t) = -\frac{dR_{T|\beta}(t)}{dt} = \frac{\omega}{\sqrt{2\pi\sigma^2\Lambda^3(t)}}\exp\left(-\frac{(\omega - \beta\Lambda(t))^2}{2\sigma^2\Lambda(t)}\right)\frac{d\Lambda(t)}{dt} = \frac{\omega}{\sqrt{\sigma^2\Lambda^3(t)}}\phi\left(\frac{\omega - \beta\Lambda(t)}{\sigma\sqrt{\Lambda(t)}}\right)\frac{d\Lambda(t)}{dt}, \tag{4}$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ denote the PDF and cumulative distribution function (CDF) of the standard normal distribution respectively. Nevertheless, the above results do not consider the influence of the random effect β on the life distribution. As for the degradation process with the measurement errors, the failure threshold ω is often defined by the industrial standard, which depends on the knowledge and experience of the expert by using the observed degradation data [13]. In this case, the life T_e is defined as

$$T_e = \inf\{t : Y(t) \geq \omega | Y(0) < \omega\} = \inf\{t : X(t) \geq \omega_e | X(0) < \omega_e\}, \tag{5}$$

where $\omega_e = \omega - \varepsilon$ is normally distributed with mean ω and standard deviation σ_ε. It becomes obvious that the life T_e is calculated by the time of $Y(t)$ first hitting the threshold ω, which is different from the life T. If ω_e and β are constant values, we can easily obtain the PDF $f_{T_e|\beta,\omega_e}(t)$ of the life T_e and the corresponding reliability function $R_{T_e|\beta,\omega_e}(t)$ by replacing ω with ω_e in (3) and (4), respectively. However, both β and ω_e are random variables instead of two constants. Therefore, the effect of β and ω_e on the life distribution should be considered simultaneously. According to the law of total probability, the unconditional PDF and reliability function of T_e can be derived respectively by

$$f(t) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f_{T_e|\beta,\omega_e}(t)p(\omega_e)p(\beta)d\omega_e d\beta = E_\beta\left[E_{\omega_e}[f_{T_e|\beta,\omega_e}(t)]\right] \tag{6}$$

and

$$R(t) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} R_{T_e|\beta,\omega_e}(t)p(\omega_e)p(\beta)d\omega_e d\beta = E_\beta\left[E_{\omega_e}[R_{T_e|\beta,\omega_e}(t)]\right], \tag{7}$$

where $p(\beta)$ and $p(\omega_e)$ represent the PDF of β and ω_e respectively, $E_\beta[\cdot]$ and $E_{\omega_e}[\cdot]$ are the expectation operators with respect to β and ω_e respectively. In the following, we attempt to achieve the exact and closed-form expression of the life distribution and reliability function in terms of the definition given in (5). In order to simplify the derivation process, we firstly provide the following lemmas and theorems.

Lemma 1. *[20] If $a_1, a_2 \in R$, then the following results hold:*

$$\int \phi(a_1 + a_2x)\phi(x)dx = \frac{1}{\sqrt{1 + a_2^2}}\phi\left(\frac{a_1}{\sqrt{1 + a_2^2}}\right)\Phi\left(x\sqrt{1 + a_2^2} + \frac{a_1a_2}{\sqrt{1 + a_2^2}}\right) + C \tag{8}$$

and

$$\begin{aligned}\int x\phi(a_1 + a_2x)\phi(x)dx = &-\frac{1}{1 + a_2^2}\phi\left(\frac{a_1}{\sqrt{1 + a_2^2}}\right)\phi\left(x\sqrt{1 + a_2^2} + \frac{a_1a_2}{\sqrt{1 + a_2^2}}\right)\\ &-\frac{a_1a_2}{(1 + a_2^2)^{3/2}}\phi\left(\frac{a_1}{\sqrt{1 + a_2^2}}\right)\Phi\left(x\sqrt{1 + a_2^2} + \frac{a_1a_2}{\sqrt{1 + a_2^2}}\right) + C,\end{aligned} \tag{9}$$

where C represents a generic constant that can change from line to line in this paper.

Theorem 1. *If $Z \sim N(\mu, \gamma^2)$ and $b_1, b_2, b_3, b_4 \in R$, then the following result holds:*

$$E_z[(b_1 + b_2Z)\phi(b_3 + b_4Z)] = \left(\frac{b_1 + b_2\mu}{\sqrt{1 + b_4^2\gamma^2}} - \frac{b_2b_4\gamma^2(b_3 + b_4\mu)}{(1 + b_4^2\gamma^2)^{3/2}}\right)\phi\left(\frac{b_3 + b_4\mu}{\sqrt{1 + b_4^2\gamma^2}}\right). \tag{10}$$

Proof. According to the results of Lemma 1 and some algebraic manipulations, we can obtain

$$\begin{aligned} E_z[(b_1+b_2Z)\phi(b_3+b_4Z)] &= \frac{1}{\gamma}\int_{-\infty}^{+\infty}(b_1+b_2z)\phi(b_3+b_4z)\phi\left(\frac{z-\mu}{\gamma}\right)dz \\ &= \int_{-\infty}^{+\infty}(b_1+b_2\mu+b_2\gamma y)\phi(b_3+b_4\mu+b_4\gamma y)\phi(y)dy \\ &= \left(\frac{b_1+b_2\mu}{\sqrt{1+b_4^2\gamma^2}}-\frac{b_2b_4\gamma^2(b_3+b_4\mu)}{(1+b_4^2\gamma^2)^{3/2}}\right)\phi\left(\frac{b_3+b_4\mu}{\sqrt{1+b_4^2\gamma^2}}\right). \end{aligned}$$

□

Lemma 2. [13] *If $Z\sim N(0,1)$ and $c_1, c_2 \in R$, then the following formula holds:*

$$E_z[\Phi(c_1+c_2Z)] = \Phi\left(\frac{c_1}{\sqrt{1+c_2^2}}\right). \tag{11}$$

Theorem 2. *If $Z\sim N(\mu,\gamma^2)$ and $d_1, d_2, d_3, d_4 \in R$, then the following result holds:*

$$E_z\Big[\exp(d_1Z+d_2Z^2)\Phi(d_3+d_4Z)\Big] = k_1^{-\frac{1}{2}}\exp\left(d_1\mu+d_2\mu^2+\frac{k_2^2\gamma^2}{2k_1}\right)\Phi\left(\frac{(d_3+d_4\mu)k_1+d_4\gamma^2k_2}{\sqrt{k_1^2+d_4^2\gamma^2k_1}}\right) \tag{12}$$

where $k_1 = 1-2d_2\gamma^2$ and $k_2 = d_1+2d_2\mu$.

Proof. According to the results of Lemma 2 and some algebraic manipulations, we have

$$\begin{aligned} E_z\Big[\exp(d_1Z+d_2Z^2)\Phi(d_3+d_4Z)\Big] &= \frac{1}{\gamma}\int_{-\infty}^{+\infty}\exp(d_1z+d_2z^2)\Phi(d_3+d_4z)\phi\left(\frac{z-\mu}{\gamma}\right)dz \\ &= \int_{-\infty}^{+\infty}\exp\big(d_1\mu+d_1\gamma x+d_2(\mu+\gamma x)^2\big)\Phi(d_3+d_4\mu+d_4\gamma x)\phi(x)dx \\ &= \exp(d_1\mu+d_2\mu^2)\int_{-\infty}^{+\infty}\exp\big(d_2\gamma^2x^2+k_2\gamma x\big)\phi(x)\Phi(d_3+d_4\mu+d_4\gamma x)dx \\ &= \exp\left(d_1\mu+d_2\mu^2+\frac{k_2^2\gamma^2}{2k_1}\right)\int_{-\infty}^{+\infty}\phi\left(\sqrt{k_1}x-\frac{k_2\gamma}{\sqrt{k_1}}\right)\Phi(d_3+d_4\mu+d_4\gamma x)dx \\ &= k_1^{-\frac{1}{2}}\exp\left(d_1\mu+d_2\mu^2+\frac{k_2^2\gamma^2}{2k_1}\right)\int_{-\infty}^{+\infty}\phi(y)\Phi\left(d_3+d_4\mu+\frac{d_4\gamma^2k_2}{k_1}+\frac{d_4\gamma}{\sqrt{k_1}}y\right)dy \\ &= k_1^{-\frac{1}{2}}\exp\left(d_1\mu+d_2\mu^2+\frac{k_2^2\gamma^2}{2k_1}\right)\Phi\left(\frac{(d_3+d_4\mu)k_1+d_4\gamma^2k_2}{\sqrt{k_1^2+d_4^2\gamma^2k_1}}\right). \end{aligned}$$

□

In light of the lemmas and theorems above, we can easily calculate (6) and (7) explicitly. The main results are summarized in the following theorem.

Theorem 3. *Given $\beta\sim N(\mu_\beta,\sigma_\beta^2)$ and $\omega_e\sim N(\omega,\sigma_\varepsilon^2)$, the PDF of the life T_e can be formulated as*

$$f(t) = \left[\frac{\omega\sigma^2+\mu_\beta\sigma_\varepsilon^2}{\zeta\sqrt{\zeta+\sigma_\beta^2\Lambda^2(t)}}-\frac{\sigma_\beta^2\sigma_\varepsilon^2\Lambda(t)(\omega-\mu_\beta\Lambda(t))}{\zeta(\zeta+\sigma_\beta^2\Lambda^2(t))^{3/2}}\right]\phi\left(\frac{\omega-\mu_\beta\Lambda(t)}{\sqrt{\zeta+\sigma_\beta^2\Lambda^2(t)}}\right)\frac{d\Lambda(t)}{dt} \tag{13}$$

where $\zeta=\sigma_\varepsilon^2+\sigma^2\Lambda(t)$.

Proof. By means of replacing ω with ω_e in (4), in the case of $\omega_e\sim N(\omega,\sigma_\varepsilon^2)$, we have

$$E_{\omega_e}[f_{T_e|\beta,\omega_e}(t)] = E_{\omega_e}\left[\frac{\omega_e}{\sqrt{\sigma^2\Lambda^3(t)}}\phi\left(\frac{\omega_e-\beta\Lambda(t)}{\sigma\sqrt{\Lambda(t)}}\right)\frac{d\Lambda(t)}{dt}\right] = \frac{\omega\sigma^2+\beta\sigma_\varepsilon^2}{\sqrt{\zeta^3}}\phi\left(\frac{\omega-\beta\Lambda(t)}{\sqrt{\zeta}}\right)\frac{d\Lambda(t)}{dt}. \tag{14}$$

In the derivation process above, based on the result of Theorem 1, it is easy to complete the proof by setting $b_1=0$, $b_2=1/\sqrt{\sigma^2\Lambda^3(t)}$, $b_3=-\beta\Lambda(t)/\sqrt{\sigma^2\Lambda(t)}$ and $b_4=1/\sqrt{\sigma^2\Lambda(t)}$ in (10). Furthermore, taking account of the randomness of the drift parameter β with $\beta \sim N(\mu_\beta, \sigma_\beta^2)$, we establish

$$f(t) = E_\beta\left[\frac{\omega\sigma^2 + \beta\sigma_\varepsilon^2}{\sqrt{\zeta^3}}\phi\left(\frac{\omega - \beta\Lambda(t)}{\sqrt{\zeta}}\right)\frac{d\Lambda(t)}{dt}\right]$$
$$= \left[\frac{\omega\sigma^2 + \mu_\beta\sigma_\varepsilon^2}{\zeta\sqrt{\zeta + \sigma_\beta^2\Lambda^2(t)}} - \frac{\sigma_\beta^2\sigma_\varepsilon^2\Lambda(t)(\omega - \mu_\beta\Lambda(t))}{\zeta(\zeta + \sigma_\beta^2\Lambda^2(t))^{3/2}}\right]\phi\left(\frac{\omega - \mu_\beta\Lambda(t)}{\sqrt{\zeta + \sigma_\beta^2\Lambda^2(t)}}\right)\frac{d\Lambda(t)}{dt}. \tag{15}$$

According to the result of Theorem 1, it is easy to complete the proof by setting $b_1 = \omega\sigma^2/\sqrt{\zeta^3}$, $b_2 = \sigma_\varepsilon^2/\sqrt{\zeta^3}$, $b_3 = \omega/\sqrt{\zeta}$ and $b_4 = -\Lambda(t)/\sqrt{\zeta}$. □

Theorem 4. *Given $\beta \sim N(\mu_\beta, \sigma_\beta^2)$ and $\omega_e \sim N(\omega, \sigma_\varepsilon^2)$, the reliability function can be expressed by*

$$R(t) = \Phi\left(\frac{\omega - \mu_\beta\Lambda(t)}{\sqrt{\zeta + \sigma_\beta^2\Lambda^2(t)}}\right) - \frac{\sigma^2}{\sqrt{\varrho}}\exp\left(\frac{2\mu_\beta\omega}{\sigma^2} + \frac{2{\mu_\beta}^2\sigma_\varepsilon^2}{\sigma^4} + \frac{2\sigma_\beta^2\kappa^2}{\sigma^4\varrho}\right)\Phi\left(-\frac{\omega(\sigma^2 + 2\sigma_\beta^2\Lambda(t)) + \mu_\beta\vartheta}{\sqrt{(\zeta + \sigma_\beta^2\Lambda^2(t))\varrho}}\right), \tag{16}$$

where $\varrho = \sigma^4 - 4\sigma_\beta^2\sigma_\varepsilon^2$, $\kappa = \omega\sigma^2 + 2\mu_\beta\sigma_\varepsilon^2$ and $\vartheta = \sigma^2\Lambda(t) + 2\sigma_\varepsilon^2$.

Proof. By means of replacing ω with ω_e in (3), we have

$$R(t) = E_\beta\left[E_{\omega_e}[R_{T_e|\beta,\omega_e}(t)]\right] = E_\beta\left[E_{\omega_e}\left[\Phi\left(\frac{\omega_e - \beta\Lambda(t)}{\sigma\sqrt{\Lambda(t)}}\right) - \exp\left(\frac{2\beta\omega_e}{\sigma^2}\right)\Phi\left(-\frac{\omega_e + \beta\Lambda(t)}{\sqrt{\sigma^2\Lambda(t)}}\right)\right]\right]$$
$$= E_\beta\left[E_{\omega_e}\left[\Phi\left(\frac{\omega_e - \beta\Lambda(t)}{\sigma\sqrt{\Lambda(t)}}\right)\right]\right] - E_\beta\left[E_{\omega_e}\left[\exp\left(\frac{2\beta\omega_e}{\sigma^2}\right)\Phi\left(-\frac{\omega_e + \beta\Lambda(t)}{\sqrt{\sigma^2\Lambda(t)}}\right)\right]\right]. \tag{17}$$

□

Based on the result of Theorem 2, setting $d_1 = d_2 = 0, d_3 = -\beta\Lambda(t)/\sqrt{\sigma^2\Lambda(t)}$ and $d_4 = 1/\sqrt{\sigma^2\Lambda(t)}$ in the case of $\omega_e \sim N(\omega, \sigma_\varepsilon^2)$, we establish

$$E_{\omega_e}\left[\Phi\left(\frac{\omega_e - \beta\Lambda(t)}{\sigma\sqrt{\Lambda(t)}}\right)\right] = \Phi\left(\frac{\omega - \beta\Lambda(t)}{\sqrt{\zeta}}\right). \tag{18}$$

Similarly, setting $d_1 = d_2 = 0, d_3 = \omega/\sqrt{\zeta}$ and $d_4 = -\Lambda(t)/\sqrt{\zeta}$ in the case of $\beta \sim N(\mu_\beta, \sigma_\beta^2)$, we obtain

$$E_\beta\left[\Phi\left(\frac{\omega - \beta\Lambda(t)}{\sqrt{\zeta}}\right)\right] = \Phi\left(\frac{\mu_\beta\Lambda(t) - \omega}{\sqrt{\zeta + \sigma_\beta^2\Lambda^2(t)}}\right). \tag{19}$$

In view of the result of Theorem 2, setting $d_1 = 2\beta/\sigma^2$, $d_2 = 0$, $d_3 = -\beta\Lambda(t)/\sqrt{\sigma^2\Lambda(t)}$ and $d_4 = -1/\sqrt{\sigma^2\Lambda(t)}$ in the case of $\omega_e \sim N(\omega, \sigma_\varepsilon^2)$, we have

$$E_{\omega_e}\left[\exp\left(\frac{2\beta\omega_e}{\sigma^2}\right)\Phi\left(-\frac{\omega_e + \beta\Lambda(t)}{\sqrt{\sigma^2\Lambda(t)}}\right)\right] = \exp\left(\frac{2\beta\omega}{\sigma^2} + \frac{2\beta^2\sigma_\varepsilon^2}{\sigma^4}\right)\Phi\left(-\frac{\omega\sigma^2 + \vartheta\beta}{\sigma^2\sqrt{\zeta}}\right). \tag{20}$$

Similarly, setting $d_1 = 2\omega/\sigma^2$, $d_2 = 2\sigma_\varepsilon^2/\sigma^4$, $d_3 = -\omega/\sqrt{\zeta}$ and $d_4 = -\vartheta/\sqrt{\sigma^4\zeta}$ in $\beta \sim N(\mu_\beta, \sigma_\beta^2)$, we obtain

$$E_\beta\left[\exp\left(\frac{2\beta\omega}{\sigma^2} + \frac{2\beta^2\sigma_\varepsilon^2}{\sigma^4}\right)\Phi\left(-\frac{\omega\sigma^2 + \vartheta\beta}{\sigma^2\sqrt{\zeta}}\right)\right] = \frac{\sigma^2}{\sqrt{\varrho}}\exp\left(\frac{2\mu_\beta\omega}{\sigma^2} + \frac{2{\mu_\beta}^2\sigma_\varepsilon^2}{\sigma^4} + \frac{2\sigma_\beta^2\kappa^2}{\sigma^4\varrho}\right)$$
$$\Phi\left(-\frac{\omega(\sigma^2 + 2\sigma_\beta^2\Lambda(t)) + \mu_\beta\vartheta}{\sqrt{(\zeta + \sigma_\beta^2\Lambda^2(t))\varrho}}\right). \tag{21}$$

So far, the proof of Theorem 4 has been completed.

Remark 1. In fact, our approach includes several special cases considering in the existing works. Firstly, our results in the case of $\Lambda(t) = t$ come down to the results obtained by Si et al. [13] when the uncertainty of the drift parameter β is not taken into account. Secondly, our results in the case of $\Lambda(t)=t$ come down to the results obtained by Peng and Tseng [12] without considering the measurement errors. Finally, if both the uncertainty of β and measurement errors are not considered, i.e., $\sigma_\varepsilon^2=\sigma_\beta^2=0$ and $\mu_\beta=\beta$, Eqs. (13) and (16) will come down to Eqs. (4) and (3) respectively. Especially in the case of $\Lambda(t)=t$, our results will be reduced to the classical inverse Gaussian distribution [21].

3. Statistical inference

Assume that the degradation processes of n independently tested units are inspected at ordered inspection times $t_1,\ldots,t_m$ with the degradation observations $\{Y_i(t_j)=y_{i,j}, i=1,\ldots,n, j=1,\ldots,m\}$, and the degradation observations among diverse units are statistically independent due to the physical independence of the tested units. Based on (1), the observed degradation process for the ith unit can be expressed as $Y_i(t_j)=\beta_i\Lambda(t_j)+\sigma W(\Lambda(t_j))+\varepsilon_{i,j}$, where β_i denotes the drift parameter for the ith unit, which is independent and identically distributed with β, and the measurement errors $\varepsilon_{i,j}$ are assumed to be independent and identically distributed realizations of ε. Define $\Delta y_{i,1}=y_{i,1}, \lambda_1=\Lambda(t_1)$, and $\Delta y_{i,j}=y_{i,j}-y_{i,j-1}, \lambda_j=\Lambda(t_j)-\Lambda(t_{j-1})$ $(j=2,3,\ldots,m)$ for the ith unit. For the sake of convenience, the first differences of the degradation observations for the ith unit is denoted by $\Delta\mathbf{Y}_i=\{\Delta y_{i,1},\ldots,\Delta y_{i,m}\}$ and the first differences of the degradation observations for all the n units are given by $\Delta\mathbf{Y}=\{\Delta\mathbf{Y}_1,\ldots,\Delta\mathbf{Y}_n\}$. For the fixed parameter β_i, based on the property of Wiener processes, $\Delta\mathbf{Y}_i$ follows a multivariate normal distribution, i.e. $\Delta\mathbf{Y}_i\sim N(\beta_i\boldsymbol{\lambda},\ \Sigma)$ with a joint PDF as follows

$$f(\Delta\mathbf{Y}_i|\beta_i)=(2\pi)^{-\frac{m}{2}}|\Sigma|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}(\Delta\mathbf{y}_i-\beta_i\lambda)'\Sigma^{-1}(\Delta\mathbf{y}_i-\beta_i\lambda)\right), \tag{22}$$

where $\Delta\mathbf{y}_i=(\Delta y_{i,1},\ldots,\Delta y_{i,m})^T$, $\lambda=(\lambda_1,\ldots,\lambda_m)^T$ and the covariance matrix Σ is positive definite matrix with

$$\Sigma_{i,j}=\begin{cases}\sigma^2\lambda_i+\sigma_\varepsilon^2, & i=j=1,\\ \sigma^2\lambda_i+2\sigma_\varepsilon^2, & i=j>1,\\ -\sigma_\varepsilon^2, & |i-j|=1,\\ 0, & otherwise.\end{cases}$$

For ease of the statistical inference, the unknown parameters are re-parameterized by $\psi=\sigma_\varepsilon^2/\sigma^2$ and $\Upsilon(\psi,\eta)=\Sigma/\sigma^2$. Let $\Theta=(\mu_\beta,\sigma_\beta^2,\sigma^2,\psi,\eta)$ be the vector of all unknown parameters, where η is the parameter in $\Lambda(t)$, and denote $\Omega=\{\beta_1,\ldots,\beta_n\}$ for notation simplicity. Therefore, given the complete data including $\Delta\mathbf{Y}$ and Ω, the complete log-likelihood function can be written as

$$\begin{aligned}\mathcal{L}(\Theta|\Delta\mathbf{Y},\Omega)&=\sum_{i=1}^{n}\Big[\ln f(\Delta\mathbf{Y}_i|\beta_i)+\ln f(\beta_i)\Big]=C-\frac{n}{2}\ln|\Upsilon(\psi,\eta)|-\frac{nm}{2}\ln\sigma^2-\frac{n}{2}\ln\sigma_\beta^2\\&\quad-\frac{1}{2}\sum_{i=1}^{n}\left[\frac{(\Delta\mathbf{y}_i-\beta_i\lambda)^T\Upsilon^{-1}(\psi,\eta)(\Delta\mathbf{y}_i-\beta_i\lambda)}{\sigma^2}+\frac{(\beta_i-\mu_\beta)^2}{\sigma_\beta^2}\right].\end{aligned} \tag{23}$$

The estimators obtained from direct constrained optimization of (23) are unsatisfactory due to the unobservability of β_i. For this reason, the EM algorithm proposed by Dempster in [22] is used to estimate the unknown parameters in this paper. The EM algorithm consists of two steps: the expectation-step (E-step) and the maximization step (M-step). The E-step computes the expectation of the log-likelihood function associated with the hidden variables, and then the M-step determines the maximizer of this expected likelihood. The two steps are repeated iteratively until the satisfactory convergence is reached. To calculate the expectation of the log-likelihood function (23), we need to obtain the first and second moments of β_i conditional on $\Delta\mathbf{Y}_i$. It is easily found that the posterior distribution of β_i conditional on $\Delta\mathbf{Y}_i$ is still the normal distribution in the case of $\beta_i\sim N(\mu_\beta,\sigma_\beta^2)$. According the Bayesian theorem, the posterior distribution of β_i can be obtained as follows

$$\begin{aligned}p(\beta_i|\Delta\mathbf{Y}_i)&\propto p(\Delta\mathbf{Y}_i|\beta_i)p(\beta_i)\\&\propto\exp\left[-\frac{1}{2\sigma^2}(\Delta\mathbf{y}_i-\beta_i\lambda)^T\Upsilon^{-1}(\psi,\eta)(\Delta\mathbf{y}_i-\beta_i\lambda)\right]\exp\left[-\frac{(\beta_i-\mu_\beta)^2}{2\sigma_\beta^2}\right]\\&\propto\exp\left\{-\frac{1}{2}\left[\left(\frac{\lambda^T\Upsilon^{-1}(\psi,\eta)\lambda}{\sigma^2}+\frac{1}{\sigma_\beta^2}\right)\beta_i^2-2\left(\frac{\Delta\mathbf{y}_i^T\Upsilon^{-1}(\psi,\eta)\lambda}{\sigma^2}+\frac{\mu_\beta}{\sigma_\beta^2}\right)\beta_i\right]\right\}\\&\propto\exp\left\{-\frac{\left[\beta_i-(\Delta\mathbf{y}_i^T\Upsilon^{-1}(\psi,\eta)\lambda\sigma_\beta^2+\mu_\beta\sigma^2)/(\lambda^T\Upsilon^{-1}(\psi,\eta)\lambda\sigma_\beta^2+\sigma^2)\right]^2}{2\sigma^2\sigma_\beta^2/(\lambda^T\Upsilon^{-1}(\psi,\eta)\lambda\sigma_\beta^2+\sigma^2)}\right\}\\&\sim N(\mu_i,\sigma_i^2)\end{aligned}$$

with

$$\mu_i=\frac{\Delta\mathbf{y}_i^T\Upsilon^{-1}(\psi,\eta)\lambda\sigma_\beta^2+\mu_\beta\sigma^2}{\lambda^T\Upsilon^{-1}(\psi,\eta)\lambda\sigma_\beta^2+\sigma^2} \tag{24}$$

and

$$\sigma_i^2 = \frac{\sigma^2\sigma_\beta^2}{\lambda^T\Upsilon^{-1}(\psi,\eta)\lambda\sigma_\beta^2+\sigma^2}. \tag{25}$$

Obviously, the second moment of β_i conditional on $\Delta\mathbf{Y}_i$ can be easily obtained by

$$u_i = E(\beta_i^2|\Delta\mathbf{Y}_i) = D(\beta_i|\Delta\mathbf{Y}_i)+[E(\beta_i|\Delta\mathbf{Y}_i)]^2 = \sigma_i^2+\mu_i^2. \tag{26}$$

In the following, we use the EM algorithm to find the MLE of Θ iteratively. Let $\Theta^{(k)} = \{\mu_\beta^{(k)}, \sigma_\beta^{2(k)}, \sigma^{2(k)}, \psi^{(k)}, \eta^{(k)}\}$ be the estimation of Θ in the kth step. To be specific, the E-step computes the expectation $\mathcal{Q}(\Theta|\Delta\mathbf{Y},\Theta^{(k)})$ of $\mathcal{L}(\Theta|\Delta\mathbf{Y},\Omega)$ with respect to Ω, we have

$$\begin{aligned}\mathcal{Q}(\Theta|\Delta\mathbf{Y},\Theta^{(k)}) &= E_{\Omega|\Theta^{(k)}}[\mathcal{L}(\Theta|\Delta\mathbf{Y},\Omega)]\\ &= C-\frac{1}{2}\sum_{i=1}^{n}\left[\frac{u_i-2\mu_i\mu_\beta+\mu_\beta^2}{\sigma_\beta^2}+\ln\sigma_\beta^2+2\ln\Phi\left(\frac{\mu_\beta}{\sigma_\beta}\right)\right]\\ &\quad -\frac{1}{2}\sum_{i=1}^{n}\left[\frac{\Delta\mathbf{y}_i^T\Upsilon^{-1}(\psi,\eta)\Delta\mathbf{y}_i-2\mu_i\Delta\mathbf{y}_i^T\Upsilon^{-1}(\psi,\eta)\lambda+u_i\lambda^T\Upsilon^{-1}(\psi,\eta)\lambda}{\sigma^2}+\ln|\Upsilon(\psi,\eta)|+m\ln\sigma^2\right],\end{aligned} \tag{27}$$

where the first term only depends on $\mu_\beta, \sigma_\beta^2$, and the second term is only associated with σ^2, ψ, η. For this reason, we can obtain the updated parameter estimates in the M-step, respectively. To be specific, for the first term, by taking the first partial derivatives of $\mathcal{Q}(\Theta|\Delta\mathbf{Y},\Theta^{(k)})$ with regard to μ_β and σ_β respectively, and setting each derivative to be zero, we have

$$\mu_\beta=\frac{1}{n}\sum_{i=1}^{n}\mu_i,\quad \sigma_\beta^2=\frac{1}{n}\sum_{i=1}^{n}u_i-\mu_\beta^2. \tag{28}$$

The estimates of $\mu_\beta^{(k+1)}$ and $\sigma_\beta^{2(k+1)}$ are obtained by (28). Similarly, for the second term, setting $\partial\mathcal{Q}(\Theta|\Delta\mathbf{Y},\Theta^{(k)})/\partial\sigma^2=0$, we can obtain the restricted estimate of $\sigma^{2(k+1)}$ as follows

$$\sigma^{2(k+1)} = \frac{1}{nm}\sum_{i=1}^{n}\left(\Delta\mathbf{y}_i^T\Upsilon^{-1}(\psi,\eta)\Delta\mathbf{y}_i-2\mu_i\Delta\mathbf{y}_i^T\Upsilon^{-1}(\psi,\eta)\lambda+u_i\lambda^T\Upsilon^{-1}(\psi,\eta)\lambda\right). \tag{29}$$

Note that $\sigma^{2(k+1)}$ depends on the estimates of ψ and η. In the following, by substituting $\mu_\beta^{(k+1)}$, $\sigma_\beta^{2(k+1)}$ and $\sigma^{2(k+1)}$ into (27), we obtain the profile likelihood function as follows

$$\begin{aligned}\mathcal{L}(\psi,\eta|\Theta^{(k)}) = C-\frac{1}{2}\sum_{i=1}^{n}\Bigg[&\ln|\Upsilon(\psi,\eta)|\\ &+m\ln\left(\frac{1}{nm}\sum_{i=1}^{n}\left(\Delta\mathbf{y}_i^T\Upsilon^{-1}(\psi,\eta)\Delta\mathbf{y}_i-2\mu_i\Delta\mathbf{y}_i^T\Upsilon^{-1}(\psi,\eta)\lambda+u_i\lambda^T\Upsilon^{-1}(\psi,\eta)\lambda\right)\right)\Bigg].\end{aligned} \tag{30}$$

The estimates of $\psi^{(k+1)}$ and $\eta^{(k+1)}$ can be achieved by maximizing the above profile likelihood function through a two-dimensional search. Next, by substituting $\psi^{(k+1)}$ and $\eta^{(k+1)}$ into (29), we can obtain the estimate of σ^2 in the $k+1$th step. In addition, the MLE of σ_ε^2 can be obtained by inverting the relation $\psi=\sigma_\varepsilon^2/\sigma^2$. The iterations are often terminated when the relative change of the expected complete log-likelihood function falls below a predefined threshold, or when the number of iterations reaches a fixed threshold.

4. Illustrative examples

4.1. Simulated example

A comprehensive Monte Carlo simulation study is firstly provided to validate the effectiveness of the statistical inference procedure based on the EM algorithm. In the simulation, the parametric form of $\Lambda(t)$ is assumed to be the power law function, i.e., $\Lambda(t)=t^\eta$, where η is a parameter to estimate. Without loss of generality, we suppose that a set of degradation paths are generated from the proposed degradation model with the following parameter settings: $\mu_\beta=5$, $\sigma_\beta=0.15$, $\sigma=3$, $\sigma_\varepsilon=0.1$ and $\eta=0.4$. The sample size of tested units is chosen to be $n=10$ and 20, and each unit is inspected m times with the measured frequency of 250 h where $m=5, 10, 15$ and 20.

The EM algorithm is applied to obtain the MLE of the known parameters based on the simulated degradation data for each combination of (n, m). The root mean square errors (RMSEs) and the biases for all known parameters are estimated using 1000 Monte Carlo replications. Table 1 presents the results of the known parameters for all combination of (n, m). It can be observed that the RMSEs and biases for each parameter become smaller as the number of degradation observations

Table 1
RMSEs and biases for parameter estimates using EM algorithm.

(n,m)	μ_β		σ_β		σ		σ_ε		η	
	RMSE	bias	RMSE	bias	RMSE	bias	RMSE	bias	RMSE	bias
(10,5)	0.3425	0.2375	0.0228	−0.0024	0.3257	0.2192	0.0427	0.0023	0.1653	−0.0627
(10,10)	0.2895	0.1857	0.0203	−0.0018	0.2461	0.1752	0.0283	−0.0013	0.0824	−0.0329
(10,15)	0.2217	0.1410	0.0154	−0.0011	0.1974	0.1340	0.0146	−0.0009	0.0457	−0.0128
(10,20)	0.1883	0.1093	0.0106	0.0009	0.1537	0.0669	0.0089	0.0006	0.0129	−0.0029
(20,5)	0.3133	0.2204	0.0182	−0.0017	0.3149	0.1919	0.0394	−0.0021	0.1458	−0.0526
(20,10)	0.2469	0.1786	0.0125	−0.0013	0.2597	0.1658	0.0219	0.0012	0.0752	−0.0251
(20,15)	0.1716	0.1328	0.0113	0.0010	0.1602	0.1297	0.0115	−0.0008	0.0397	−0.0108
(20,20)	0.1524	0.1054	0.0097	0.0007	0.1268	0.0524	0.0063	0.0005	0.0105	−0.0013

Table 2
Light intensity degradation data of 12 LEDs over time [23].

Time / No.	1	2	3	4	5	6	7	8	9	10	11	12
0	0	0	0	0	0	0	0	0	0	0	0	0
50	13.4	17.9	17.3	20.2	24.9	16.3	27.0	13.8	18.8	33.2	33.9	23.5
100	21.3	28.6	29.7	31.7	33.3	26.0	35.0	32.4	35.0	36.7	35.8	38.3
150	24.0	34.6	36.0	37.7	37.2	32.6	39.3	37.3	39.4	40.7	40.6	38.7
200	28.4	38.3	38.7	40.0	41.0	37.0	41.7	40.0	40.7	42.7	42.0	40.3
250	32.0	42.0	40.7	41.0	46.0	38.7	42.0	40.3	42.7	43.5	44.7	44.0

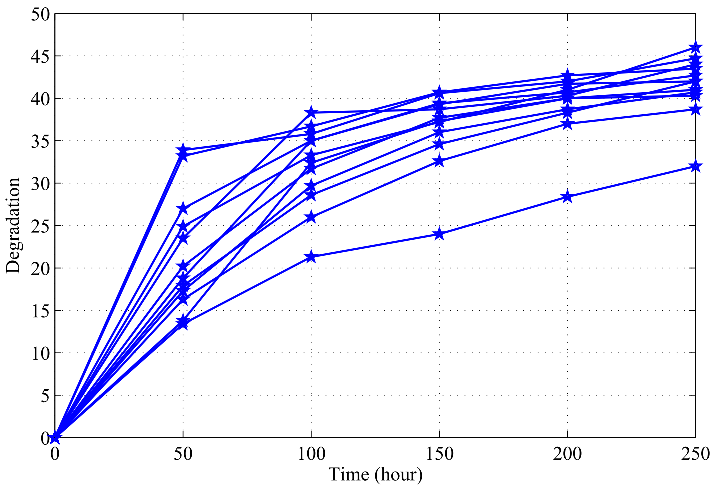

Fig. 1. The degradation paths of 12 LEDs.

increases, i.e. the estimated parameter values tend to approximate the actual values with the increasing of the available degradation data. The major reason lies in that the estimated results for each parameter can be adaptively updated by using the EM algorithm as the degradation observations increase. Therefore, it is effective to employ the MLE method based on the EM algorithm to estimate the model parameters.

4.2. Case study

In this subsection, we use the degradation data of LED lamps to demonstrate the applicability and superiority of the developed approach. The light intensity of LED lamps decreases over time during its whole life cycle. A degradation test in [23] has been performed to research the degradation behavior of LED lamps. In the degradation test, 12 units are tested under an accelerated current of 40 mA with measure frequency 50 h until 250 h passes. A LED lamp is regarded as a soft failure when the drop of its light intensity exceeds a failure threshold $\omega = 50$. Table 2 presents the degradation data of 12 tested units, and the degradation paths are also shown in Fig. 1. It can be observed that the degradation evolution of LED lamps cannot be effectively described by a linear degradation model. Therefore, a time-transformed Wiener process is adopted to describe the degradation process of LED lamps. In the following, we use the degradation data to illustrate the superiority of the presented reliability estimation approach by comparing with Ye's method only through the direct maximization of the likelihood function in [14].

Table 3
Comparisons of parameter estimation results between our method and Ye's method.

Method	$\hat{\mu}_\beta$	$\hat{\sigma}^2_\beta$	$\hat{\sigma}^2$	$\hat{\sigma}^2_\varepsilon$	$\hat{\eta}$	Log-likelihood
M_1	5.58	1.69×10^{-2}	10.59	1.98×10^{-12}	0.364	−151.4
M_2	5.44	2.65×10^{-13}	10.24	2.528×10^{-12}	0.368	−158.9

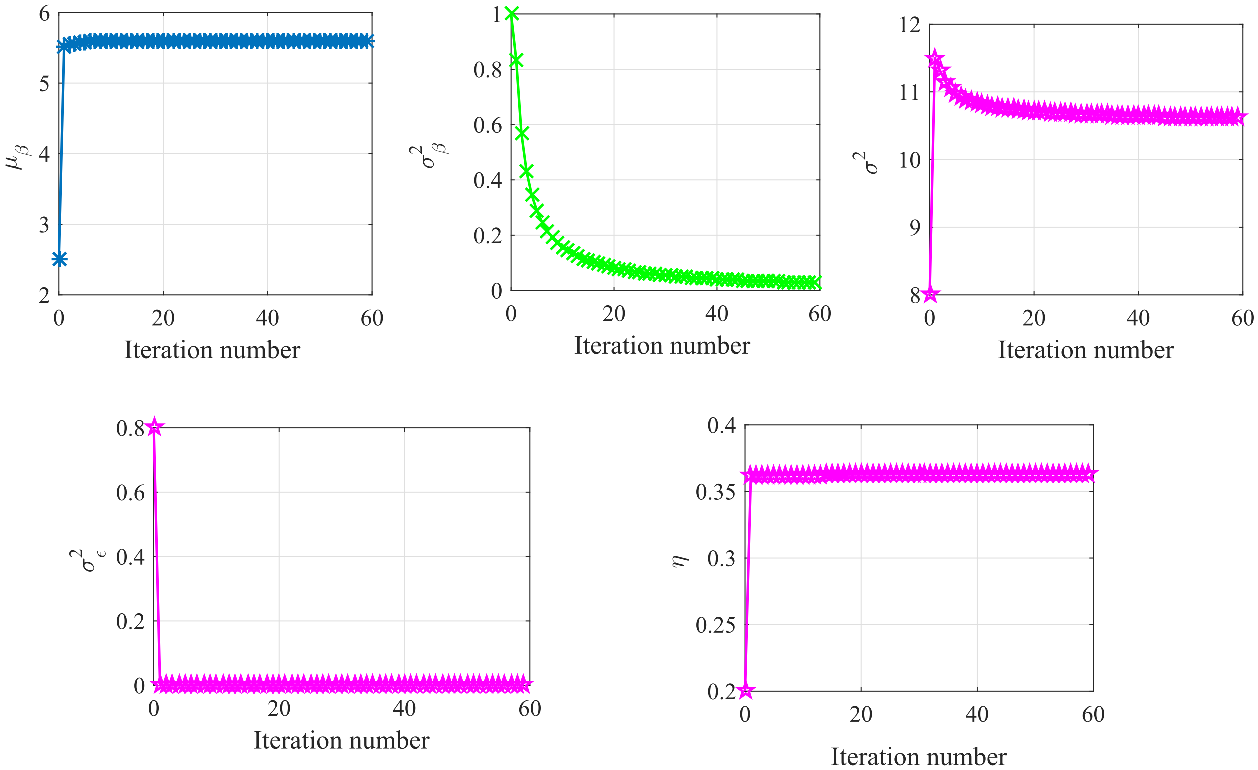

Fig. 2. Illustration of the iterative convergence based on the EM algorithm.

The empirical research works show that power law function, i.e., $\Lambda(t) = t^\eta$, is well-suited for the degradation modeling of LED lamps [14]. In the following, our method and Ye's method are denoted by M_1 and M_2 respectively for the sake of simplicity. Based on the EM algorithm presented in Section 3, we can obtain the MLE results of the model parameters, in which the prior parameters $\mu_\beta, \sigma^2_\beta, \sigma^2, \psi, \eta$ are selected at random. The estimated parameters and the associated values of the log-likelihood function are shown in Table 3. Meanwhile, Table 3 also gives the estimated results in terms of Ye's method [14], in which the appropriate choice of the initial parameters is essential.

As shown in Table 3, the obtained values of the log-likelihood function using our method is significantly smaller in comparison with Ye's method, which demonstrates our method is appropriate to fit the degradation data of LED lamps. In addition, it can be observed that the estimated value of $\hat{\sigma}^2_\beta$ using our method is significantly greater than the estimated result using Ye's method, which indicates that the randomness of β is obvious for LED degradation modeling. From Fig. 1, we can observe that the degradation paths for 12 LED lamps have significant differences, i.e., there exists obvious unit-to-unit heterogeneity. In general, the estimated results using Ye's method have larger biases because of the unobservability of β. Under the circumstance of the existence of unobserved hidden variables, however, the EM algorithm adopted in this paper can eliminate such defect and provide more accurate estimated results. In order to further verify the effectiveness of the EM algorithm, Fig. 2 illustrates the convergence of iterative sequences generated by the EM algorithm, and we can find that each model parameter can achieve a convergence, which indicates the EM algorithm has good convergence performance. Fig. 3 also compares the estimated PDF and reliability function from our method with Ye's method. It can be observed that there exists a few differences between the estimated results. The main reason lies in the fact that the EM algorithm in our work can effectively converge to the optimal solution when the likelihood function includes unobserved hidden variables, in which the initial parameters are randomly selected in parameter estimation. However, Ye's method uses the direct maximization of the log-likelihood function to estimate the model parameters, in which the appropriate selection of the initial parameters is a critical step. The inappropriate selection will lead to a very slow convergence, as well as a local optimum far away the true parameter values [14]. In addition, even if the choice of the initial parameters is appropriate, the EM algorithm has better

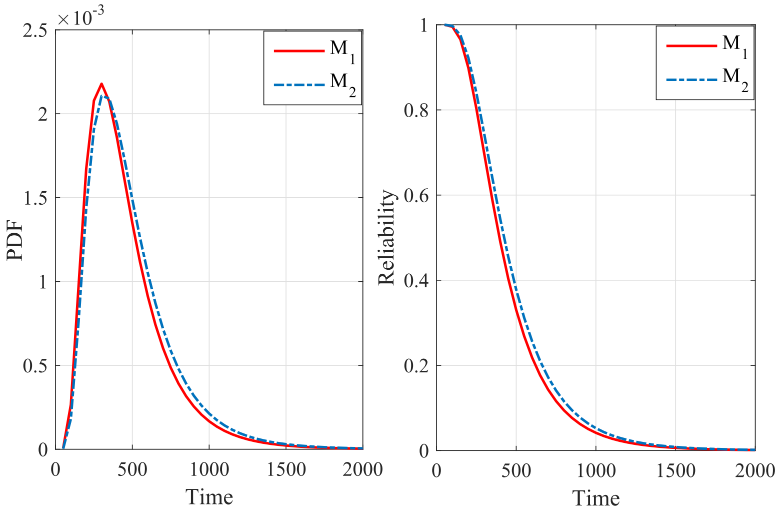

Fig. 3. Comparison of the estimated PDF and reliability function for our method and Ye's method.

convergence performance in comparison with direct maximization of the log-likelihood function. Therefore, compared with Ye's method, the proposed method can achieve more reasonable reliability estimation results, and the obtained results can be more effective to schedule maintenance actions in the stage of preventive maintenance planning.

5. Conclusions

This paper presents a reliability estimation approach based on a time-transformed Wiener process with jointly considering temporal variability, measurement errors and unit-to-unit heterogeneity. To satisfy the active requirements in applications such as maintenance scheduling and logistic support, the closed-form expressions of the PDF of the life and reliability function are also derived in detail. In the statistical inference procedure, the EM algorithm is effectively used to estimate the model parameters. By comparing with the existing approach, the MLE based on the EM algorithm can improve the efficiency of parameter estimation. The case study for LED lamps has demonstrated that the proposed approach can obtain better reliability estimation results and have great application potentials in preventive maintenance scheduling.

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